

Fig. 6.1 Percentage reabsorption of filtered NaCl along the eu-volemic nephron. ALH, Ascending limb of the loop of Henle; CCD, cortical collecting duct; DCT, distal convoluted tubule; DLH, descending thin limb of the loop of Henle; IMCD, inner medullary collecting duct; OMCD, outer medullary collecting duct; PCT, proximal convoluted tubule; PST, proximal straight tubule; TALH, thick ascending limb of the loop of Henle. (From Mos OW, Baum M, Beny CA, Rector FC Jr. Renal transport of glucose, amino acids, sodium, chloride, and water. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: WB Saunders; 2004. p. 413–432.)

Fig_6_1

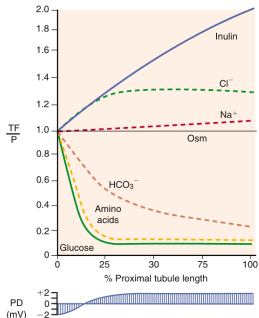


Fig. 6.2 Concentration of solutes along the proximal tubule in relation to the transepithelial potential difference (PD). Osm, Osmolality; TF/P, ratio of tubule fluid-to-plasma-concentration. (From Rector FC Jr. Sodium, bicarbonate, and chloride absorption by the PT. *Am J Physiol*. 1983;244:F461–F471.)

Fig_6_2

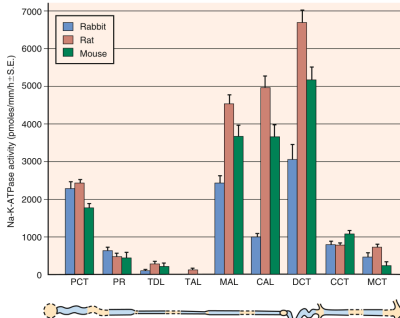


Fig. 6.3 Distribution of Na⁺-K⁺-ATPase activity along the nephron. cTAL, Cortical thick ascending limb; CCD, cortical collecting duct; DCT, distal convoluted tubule; mTAL, medullary thick ascending limb; MCD, medullary collecting duct; PCT, proximal convoluted tubule; PST, proximal straight tubule; tAL, thin ascending limb of the loop of Henle; tDL, descending thin limb of the loop of Henle. (From Katz AI, Doucet A, Morel F. Na-K-ATPase activity along the rabbit, rat, and mouse nephron. *Am J Physiol*. 1979;237:F114–F120.)

Fig_6_3

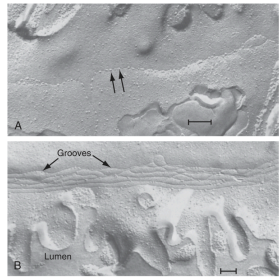


Fig. 6.4 Freeze-fracture electron microscopy images of tight junctions in mouse proximal and distal nephron. (A) Proximal convoluted tubule, a "leaky" epithelium; the tight junction contains only one junctional strand, seen as a groove in the fracture face (arrows). (B) Distal convoluted tubule, a "tight" epithelium. The tight junction is deeper and contains several anastomosing strands, seen as grooves in the fracture face. (From Claude P, Goodenough DA. Fracture faces of zonula occludens from "tight" and "leaky" epithelia. *J Cell Biol*. 1973;58:390–400.)

Fig_6_4

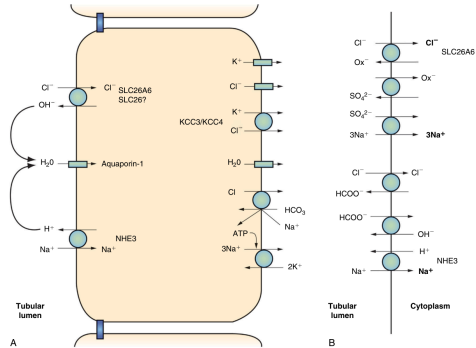


Fig. 6.5 Transapical NaCl transport in the proximal tubule. (A) In the simplest scheme, Cl⁻ enters the apical membrane via a Cl⁻-OH⁻ exchanger, coupled to Na⁺ entry via NHE3. (B) Alternative apical anion exchange activities that couple to Na⁺-H⁺ exchange and Na⁺-SO₄²⁻ cotransport. (C) Alternative apical anion exchange activities that couple to Na⁺-H⁺ and Na⁺-SO₄²⁻ cotransport. See text for details.

Fig_6_5

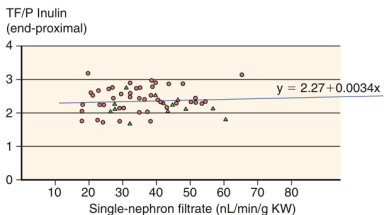


Fig. 6.6 Glomerulotubular balance. The tubular fluid-to-plasma ratio of the nonreabsorbable marker, inulin (TF/P Inulin), at the end of the proximal tubule (PT), which is used as a measure of fractional water absorption by the PT, does not change as a function of single-nephron GFR (SNGFR). Measurements were done during antidiuresis (triangles) and water diuresis (circles). (From Schnermann J, Wahl M, Liebau G, Fischbach H. Balance between tubular flow rate and net fluid reabsorption in the proximal convoluted of the rat kidney. I. Dependency of reabsorptive net fluid flux upon proximal tubular surface area at spontaneous variations of filtration rate. *Phlegus Arch*. 1968;304:90–103.)

Fig_6_6

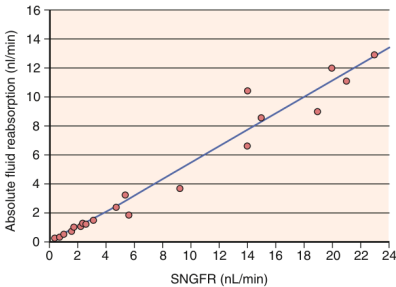


Fig. 6.7 Glomerulotubular balance. Shown is the linear increase in absolute fluid reabsorption by the late proximal tubule as a function of single-nephron GFR (SNGFR). (From Spitzer A, Brandis M. Functional and morphologic maturation of the superficial nephrons. Relationship to total kidney function. *J Clin Invest*. 1974;53:279–287.)

Fig_6_7

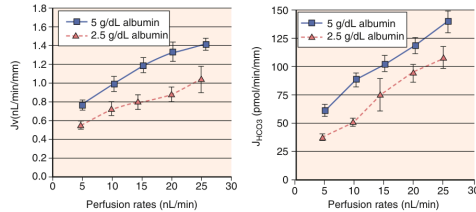


Fig. 6.8 Glomerulotubular balance; flow-dependent increases in fluid (J_v) and HCO_3^- ($J_{\text{HCO}_3^-}$) absorption by perfused mouse proximal tubules. Absorption also increases when bath albumin concentration increases from 2.5 to 5 g/dL. (From Du Z, Yan Q, Duan Y, et al. Axial flow modulates PT NHE3 and H-ATPase activities by changing microvilli bending moments. *Am J Physiol Renal Physiol*. 2006;290:F289–F296.)

Fig_6_8

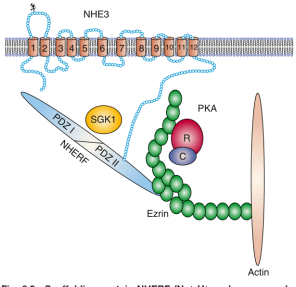
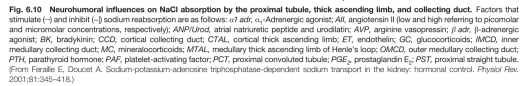


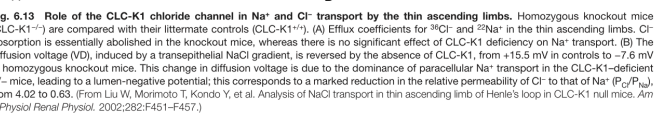
Fig. 6.9 Scaffolding protein NHERF (Na⁺-H⁺ exchanger regulatory factor) links the Na⁺-H⁺ exchanger NHE3 to the cytoskeleton and signaling proteins. NHERF binds to ezrin, which in turn links to protein kinase A (PKA) and the actin cytoskeleton. NHERF also binds to SGK1 (serum- and glucocorticoid-induced kinase 1), which activates NHE3. C, Catalytic subunit of PKA; PDZ, domain named for the PSD95, Drosophila disk large (Drosophila), and ZO-1 proteins; R, regulatory subunit of PKA.

Fig_6_9

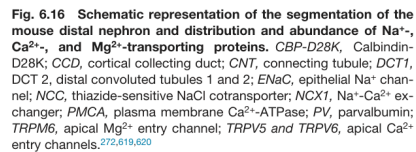
06. Transport of Sodium, Chloride, and Potassium · assets 10-18 / 25



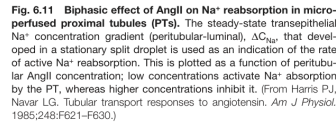
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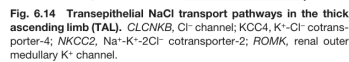
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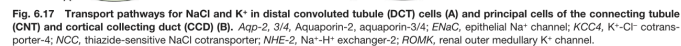
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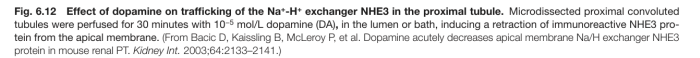
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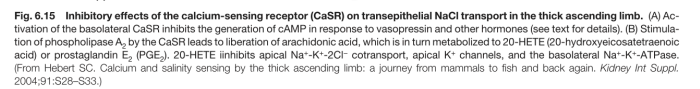
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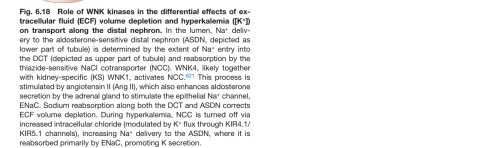
Fig_6_17



Fig_6_12



Fig_6_15



Fig_6_18

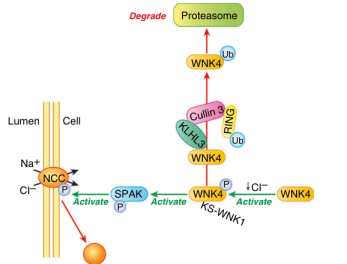


Fig. 6.19 Detailed scheme of WNK kinase regulation. NCC is phosphorylated and activated by SPAK, which is phosphorylated and activated by WNK4, which is activated by low intracellular chloride concentration. WNKs can interact with a cullin-ring ligase, via the adaptor KLHL3, leading to WNK ubiquitination and subsequent degradation by the proteasome. When dephosphorylated, NCC can be removed from the apical plasma membrane.

Fig_6_19

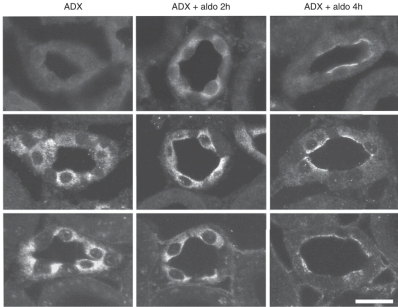


Fig. 6.22 Immunofluorescence images of connecting tubule (CNT) profiles in kidneys from adrenalectomized rats (ADX) and from ADX rats 2 and 4 h after aldosterone (aldo) injection. Antibodies against the α -, β -, and γ -subunits of ENaC reveal absent expression of the former in ADX rats, with progressive induction by aldosterone. All three subunits traffic to the apical membrane in response to aldosterone. This coincides with rapid aldosterone induction of the SGK kinase in the same cells; SGK is known to increase the expression of ENaC at the plasma membrane (see text for details). Bar = 15 μ m. [From Loffing J, Zecovic M, Fecarotta E, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280:F675–F682.]

Fig_6_22

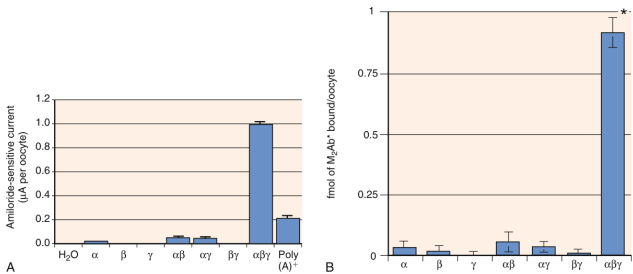


Fig. 6.20 Maximal expression of the amiloride-sensitive epithelial Na⁺ channel (ENaC) at the plasma membrane requires the co-expression of all three subunits (α -, β -, and γ -ENaC). (A) Amiloride-sensitive current in *Xenopus* oocytes expressing the individual subunits and various combinations thereof. (B) Surface expression is markedly enhanced in *Xenopus* oocytes that coexpress all three subunits. The individual cDNAs were engineered with an external epitope tag; expression of the channel proteins at the cell surface is measured by binding of a monoclonal antibody (M₂Ab) to the tag. Poly A⁺. Polyadenylated mRNA. [A from Canessa CM, Schild L, Buiel G, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367:463–467. B from Firooz D, Schild L, Gautschi I, et al. Cell surface expression of the epithelial Na channel and a mutant causing Liddle syndrome: a quantitative approach. *Proc Natl Acad Sci U S A*. 1996;93:15370–15375.]

Fig_6_20

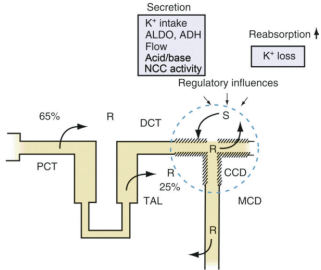


Fig. 6.23 K⁺ transport along the nephron. Approximately 90% of filtered K⁺ is reabsorbed by the PT and the loop of Henle. K⁺ is secreted along the initial and cortical collecting ducts. Net reabsorption occurs in response to K⁺ depletion, primarily within the medullary collecting duct. ADH, Antidiuretic hormone; ALDO, aldosterone; CCD, cortical collecting duct; DCT, distal convoluted tubule; MCD, medullary collecting duct; PCT, proximal convoluted tubule; R, reabsorption; S, secretion; TAL, thick ascending limb.

Fig_6_23

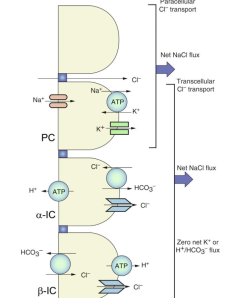


Fig. 6.21 Transapical Cl⁻ transport by principal (PC) and α -IC and β -IC intercalated cells. The lumen-negative potential difference generated by principal cells drives paracellular Cl⁻ absorption. Alternatively, transcellular transport occurs in β -intercalated cells via apical Cl⁻/HCO₃⁻ exchange (SLC26A4/pendrin) and basolateral Cl⁻ exit via ClC-4/5. Modified from Mac OSW, Baum M, Berry CA, Pector FC Jr. Renal transport of glucose, amino acids, sodium, chloride, and water. In Brenner BM, ed. *Brenner and Rector's The Kidney*. Philadelphia: WB Saunders; 2004. p. 415–462.]

Fig_6_21

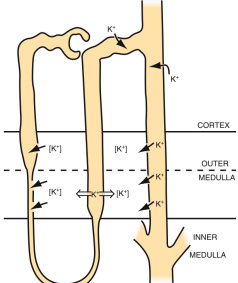


Fig. 6.24 Schematic representation of medullary K⁺ recycling. Medullary interstitial K⁺ increases considerably after dietary K⁺ loading due to the combined effects of secretion in the cortical collecting duct, absorption in the outer medullary collecting duct, thick ascending limb, and inner medullary collecting duct, and secretion in the descending thin limb. See text for details. [From Slicks JB. Consequences of potassium recycling in the renal medulla. Effects of ion transport by the medullary thick ascending limb of Henle's loop. *J Clin Invest*. 1992;90:1219–1228.]

Fig_6_24

Table 6.1 Effect of High-K⁺ Diet, Aldosterone, and/or NaCl Restriction on Small-Conductance Channel Density in the Rat Cortical Collecting Duct

Parameter	K ⁺ Channel Density (μ m ²)	Plasma Aldosterone (ng/dL)	Plasma K (mmol/L)
Control	0.41	15	3.68
High-K ⁺ diet, 6h	1.51	36	NM
High-K ⁺ diet, 48h	2.13	98	4.37
Low-Na ⁺ diet, 7 days	0.48	1260	NM
Aldosterone infusion, 48h	0.44	550	2.44
Aldosterone + high-K ⁺ diet	0.32	521	3.80

NM, Not measured.
Modified from Palmer LG, Frindt G: Regulation of apical K channels in rat cortical collecting tubule during changes in dietary K intake. *Am J Physiol* 277:F805-F812, 1999.

Table_6_1